MIL-G-0016265B(SHIPS)
30 April 1958
Used IN LIEU OF
MIL-G-16265A (In Part)*
17 April 1952

MILITARY SPECIFICATION

GASKETS, METALLIC-ASBESTOS, SPIRAL WOUND

(FOR NAVY FLANGED JOINTS IN PIPING SYSTEMS)

This limited coordination Military specification has been prepared by the Bureau of Ships based upon currently available technical information, but it has not been approved for promulgation as a revision of Military Specification MIL-G-16265A(in part). It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 Scope.* This specification covers gaskets of Navy design for use in Navy flanged joints in steam piping systems for pressures up to and including 600 pounds per square inch (p.s.i.) and temperatures up to and including 875 degrees Fahrenheit (°F.).

1.2 Classification. - Gaskets shall be of the following types and classes, as specified (see 6.2).

Type I - Pipeline flange gaskets, with outer metal ring.
Type II - Refills for type I gaskets.
Series 150.
Series 300.
Series 400.
Series 600.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL
QQ-S-763 - Steel Bars, Shapes and Forgings-Corrosion Resisting.
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Fiber.
PPP-B-676 - Boxes, Set-Up, Paperboard.
PPP-T-76 - Tape, Pressure-Sensitive Adhesive Paper, Water Resistant.

*See 6.4.

FSC 5330

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MILITARY
MIL-P-116 - Preservation, Methods of.
MIL-B-4229 - Boxes; Paperboard, Metal-Stayed.
MIL-B-10377 - Box: Wood Cleated, Veneer, Paper Overlaid.
MIL-L-10547 - Liners, Case, Waterproof.

NAVY DEPARTMENT
General Specifications for Inspection of Material.

STANDARDS

MILITARY
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

BUREAU OF SHIPS
B-106 - Flanged Cast Steel Pipe Fittings.
B-110 - Steel Flanges for Steel Tubing.
B-141 - Flanges, Steel.
B-152 - Flanged Cast Steel Pipe Fittings.
B-159 - Flanges, Steel, Spherical Back Van Stone.
B-174 - Flanges, Steel, Spherical Back Van Stone.
B-175 - Flanged Cast Steel Pipe Fittings.
B-196 - Steel Flanges for Steel Tubing, Welding Neck.
5000-S4823-64496 - Fittings, Forged Steel Socket Welding and Flanged Nipples.
5000-S4824-1385571 - Valves, Cast Steel, Flanged, Globe, Angle and Cross.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, N.Y.)

3. REQUIREMENTS

3.1 Qualification.- The gaskets furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein (see 6.3).

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3.2 Design.-

3.2.1 Type I, all series, pipeline flange gaskets with outer metal ring.- All series of gaskets shall consist of a type II refill snapped into a solid metal outer ring with a loose sliding fit. The ring shall be of soft carbon steel and shall have a thickness of 3/32 inch ±0.006 inch. The ring shall have a V-shaped groove or other satisfactory retaining feature machined on the inside perimeter to fit the outside perimeter of a type II refill. The design shall be such that when the gasket is compressed as specified in tables I and II, the load on the flange bolts or bolt studs shall equal a load equivalent to 25,000 or 30,000 as applicable, ±10 percent, p.s.i. of total cross-sectional area at the root of threads of bolt studs per flange. The gaskets shall conform to the requirements of tables I and II.

Table I - Gasket characteristics for series 150 Navy flanges - B110.[1]

Iron pipe size (i.p.s.)	Pipe outside diameter (o.d.)	Gasket refills		Minimum number of plies				Outer[3] metal ring o.d.	Test load[4] gasket pressure	Gasket[5] thickness under compression
				Metal			Asbestos			
		I.d. [2]	O.d. [2]	Inside periphery	Outside periphery	Total	Total			
Inches	Inches	inches	inches					Inches	Pounds	Inch
1/4	0.540	3/4	1- 1/8	3	2	8	4	1- 5/8	11,340	0.125
3/8	.675	7/8	1- 1/2	3	2	8	4	1- 3/4	11,340	.132
1/2	.840	1	1- 3/8	3	2	8	4	1-15/16	11,340	.132
3/4	1.050	1-1/4	1-11/16	3	2	9	5	2- 3/16	15,120	.132
1	1.315	1-1/2	2	3	2	10	6	2- 5/8	15,120	.132
1-1/4	1.660	1-3/4	2- 1/4	3	2	10	6	2- 7/8	15,120	.135
1-1/2	1.900	2-1/4	2- 3/4	3	2	10	6	3- 7/16	22,680	.135
2	2.375	2-3/4	3- 5/16	3	2	10	6	3-15/16	22,680	.135
2-1/2	2.875	3-3/8	4	3	2	12	8	4- 1/2	22,680	.135
3	3.500	3-7/8	4-1/2	3	2	12	8	5	30,240	.135
3-1/2	4.000	4-3/8	5	3	2	12	8	5- 9/16	30,240	.135
4	4.500	4-7/8	5-1/2	3	2	12	8	6- 1/16	30,240	.135
4-1/2	5.000	5-3/8	6 1/16	6	3	16	8	6- 9/16	37,800	.135
5	5.563	5-7/8	6- 5/8	6	3	16	8	7- 3/16	60,600	.135
5-1/2	6.120	6-1/2	7- 1/4	6	3	16	8	7-11/16	60,600	.135
6	6.625	7	7-13/16	6	3	17	9	8- 1/4	72,720	.135
7	7.625	8	8-13/16	6	3	17	9	9- 3/8	72,720	.135
8	8.625	9	9- 7/8	6	3	18	10	10- 7/16	84,840	.135
9	9.625	10	11- 1/8	6	3	22	14	11- 5/8	126,896	.135
10	10.750	11	12- 1/8	6	3	22	14	12-11/16	135,960	.135
11	11.750	11-1/2	12- 5/8	6	3	22	14	14- 1/4	145,024	.135
12	12.750	12-1/2	13- 3/4	6	3	25	17	15- 5/16	163,152	.135
14 o.d.	14.000							16- 1/2	238,830	.135
15 o.d.	15.000							17- 1/2	251,400	.135
16 o.d.	16.000							18- 9/16	251,400	.140
18 o.d.	18.000							20- 5/8	276,540	
20 o.d.	20.000							22-11/16	396,120	

[1]Gasket for 150 p.s.i., 850°F. steam service when used with Drawing B-110.
[2]Tolerance ±1/64 inch.
[3]I.d. of outer metal ring equals o.d. of refills plus 1/32 inch for size under 4 inches and 1/16 inch for size 4 inches and above with a tolerance of plus or minus 1/64 inch.
[4]Corresponds to bolting load of 30,000 p.s.i. unit stress. Bolts; 3/4 inch diameter, 10 threads; 7/8 inch diameter, 9 threads; all other bolts 8 threads.
[5]Tolerance plus or minus 0.005 inch.

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Table II - Series 300, 400, and 600 Navy flanges.

I.p.s.	Pipe o.d.	Gasket refills		Minimum number of plies				Outer metal ring[2] o.d. inch			Test load[6] gasket pressure	Gasket[7] thickness under compression
				Metal			Asbestos	[3]Series 300	[4]Series 400	[5]Series 600		
		I.d. inches[1]	O.d. inches[1]	Inside periphery	Outside periphery	Total	Total					
Inches	Inches										Pounds	Inch
1/4	0.540	1/2	27/32	4	3	9	3	2- 1/8	1- 3/4	1- 3/4	12,565	0.125
3/8	.675	3/4	1- 7/32	6	3	12	4	2- 3/8	2- 1/8	2- 1/8	12,565	.132
1/2	.840	3/4	1- 7/32	6	3	12	4	2- 3/8	2- 1/8	2- 1/8	12,565	.132
3/4	1.050	1	1-17/32	6	3	14	6	2-11/16	2- 5/8	2- 5/8	20,180	.132
1	1.315	1- 1/4	1-27/32	6	3	15	7	3- 1/8	2- 7/8	2- 7/8	20,180	.132
1-1/4	1.660	1-11/16	2-11/32	6	3	15	7	3- 7/16	3- 1/4	3- 1/4	24,216	.135
1-1/2	1.900	2	2-23/32	6	3	17	9	4	3- 3/4	3- 3/4	36,240	.135
2	2.375	2- 9/16	3-11/32	6	3	17	9	4- 1/2	4- 5/16	4- 5/16	48,432	.135
2-1/2	2.875	3- 1/16	3-27/32	6	3	18	10	5- 3/16	5- 1/16	5- 1/16	72,480	.135
3	3.500	3- 3/4	4-11/16	6	3	20	12	5- 3/4	5-13/16	5-13/16	72,480	.135
3-1/2	4.000	4- 1/8	5- 3/16	6	3	22	14	6- 5/16	6- 5/16	6- 5/16	100,632	.135
4	4.500	4- 3/4	5-13/16	6	3	22	14	6- 7/8	6-15/16	6-15/16	100,632	.135
4-1/2	5.000	5- 5/16	6- 7/16	6	3	24	16	7- 7/16	7- 9/16	7- 7/16	132,240	.135
5	5.563	5-13/16	6-15/16	6	3	24	16	8	8- 5/16	8- 3/16	132,240	.135
5-1/2	6.125	6- 3/8	7- 9/16	6	3	26	18	8-11/16	8-13/16	8-15/16	150,948	.135
6	6.625	6- 7/8	8- 3/16	6	3	27	19	9- 1/4	9-11/16	9- 9/16	198,360	.135
6-1/2	7.125	7- 7/16	8-11/16	6	3	27	19	9- 7/8	10- 5/16	10- 3/16	198,360	.135
7	7.625	7-11/16	9- 3/16	6	3	29	21	10- 7/16	10-13/16	10-11/16	262,008	.135
7-1/2	8.125	8- 5/16	9-13/16	6	3	29	21	11- 1/8	11- 5/16	11- 5/16	262,008	.135
8	8.625	8- 7/8	10- 5/16	6	3	30	22	11-11/16	11-15/16	11-13/16	262,008	.135
8-1/2	9.125	9- 5/16	10-13/16	6	3	32	24	12- 1/4	12- 5/16	--------	262,008	.135
9	9.625	9- 7/8	11- 7/16	6	3	34	26	12-13/16	12-13/16	13- 7/16	334,440	.135
9-1/2	10.250	10- 5/16	11-15/16	6	3	34	26	13- 3/8	13- 9/16	--------	349,344	.135
10	10.750	10-13/16	12- 7/16	6	3	35	27	13-15/16	14- 1/16	15-11/16	445,920	.135
10-1/2	11.250	11- 5/16	12-15/16	6	3	34	26	14- 1/2	14- 9/16	--------	349,344	.135
11	11.750	11- 3/4	13-11/16	6	3	37	29	15- 3/16	15- 7/16	16-11/16	557,400	.135
11-1/2	12.250	12- 5/16	14- 3/16	6	3	35	27	15- 3/4	15-15/16	--------	445,920	.135
12	12.750	12- 7/8	14-11/16	6	3	35	27	16- 3/8	16- 7/16	17-15/16	557,400	.135
14 o.d.	14.000	14- 1/4	16	6	3	32	24	17- 1/2	18-15/16	--------	557,400	.135
15 o.d.	15.000	15- 1/4	17	6	3	32	24	18- 3/4	20- 3/16	--------	557,400	.135
16 o.d.	16.000	16- 1/4	18- 1/4	6	3	36	28	19-13/16	21- 1/16	--------	577,000	.140

[1]Tolerance plus or minus 1/64 inch.
[2]I.d. of outer metal ring equals o.d. of refills plus 1/32 inch for size under 3-1/2 inches and 1/16 inch for size 3-1/2 inches and above with a tolerance of plus or minus 1/64 inch.
[3]Gaskets for 300 p.s.i. steam service when used with Drawing B-141, B-106 or 5000-S4824-1385571.
[4]Gaskets for steam service pressure 301 to 400 p.s.i. gage and 850°F. Maximum temperature and boiler feed service pressure 401 to 550 p.s.i. and 425°F. maximum temperature, when used with Drawing B-152, B-159 or 5000-S4823-64496.
[5]Gaskets for steam service pressure 401 to 600 p.s.i. gage and 850°F., maximum temperature and boiler feed service pressure 551 to 750 p.s.i. gage and 425°F., maximum temperature, when used with Drawing B-196, B-174, B-175, 5000-S4823-64496 and valves for 600-pound service.
[6]Corresponds to bolting load of 25,000 p.s.i. unit stress for 1/4 inch to 1 inch inclusive and 16 inches o.d. and 30,000 p.s.i. for 1-1/4 inch to 15 inch o.d. inclusive.
[7]Tolerance plus or minus 0.005 inch.

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3.2.2 Type II refills for all series of type I gaskets.- Refills shall be of the inside and outside dimensions as specified in tables I and II (see 6.2). These dimensions shall be in accordance with the line flange joints for which they are designed (see tables I and II). The refill shall assemble with the outer metal ring of maximum permissible inside diameter as allowed by the tolerances specified in tables I and II.

3.3 Material.-

3.3.1 Metal strip.- The metal strip used in the manufacture of the gasket refill shall conform to class 304 of Specification QQ-S-763.

3.4 Construction.- The gaskets shall be composed of a single alternate strip of metal with asbestos wound in spiral dovetail shape between some of the plies, starting on the inside and working outward ply by ply to the desired size (see note 3 of table I and note 2 of table II). The asbestos paper may be either a single strip or in two strips. Spot welds on the gasket inside diameter shall be not less than three in number and equally spaced at 2 inches maximum spacing. The outside weld shall consist of not less than three spot welds. The three or more outside spot welds shall develop the strength of the metal strip in tension.

3.4.1 Pressure.- Complete gaskets and refills shall be suitable for the nominal steam pressure specified in tables I and II, as applicable and temperature up to 850°F.

3.4.2 Metal strip.- The metal strip shall be not less than 0.007 and not more than 0.009 inch in thickness.

3.4.3 Asbestos strip.- The asbestos strip shall be not more than 0.022 inch thick.

3.4.4 Gasket thickness.- The thickness of the finished gasket shall be 0.175 inch $\pm$ 0.005 inch.

3.4.5 Plies.- A metal ply shall consist of one 360 degree turn of metal strip. Plies shall be counted adjacent to a terminal weld. An asbestos ply shall consist of one 360 degree turn of asbestos strip(s) depending upon the design.

3.4.6 Types I and II gaskets shall pass the endurance, shock and compression load tests specified in 4.5. Type II gaskets shall be temporarily snapped into outer metal rings, while undergoing tests.

3.5 Marking.- The metal ring for the type I gaskets shall be steel-stamped on one side to show the pipe size for which intended and corresponding steam pressure (see 6.2). Where refills are purchased separate from the outer rings, each refill shall have the following markings on a strip of pressure sensitive tape: Specification "MIL-G-16265", the pipe size, and the pressure series followed by a suffix letter "N" (to designate use with Navy flanges). For those sizes where the refill is identical with the refill in Specification MIL-G-21032, use the suffix "N (ASA)". (Example: Specification MIL-G-16265 - 3-600 N(ASA).) In applying the pressure sensitive tape, the tape shall be looped through the refill and brought back upon itself for firm retention.

3.6 Workmanship.- The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Qualification tests at a laboratory.- Qualification tests shall be conducted at a laboratory designated by the Bureau of Ships. These tests shall consist of the tests specified in 4.5.

4.2 Sampling for acceptance inspection.-

4.2.1 Lot.- For purposes of sampling, a lot shall consist of not more than 5000 gaskets of one type, series and size, wound on one machine by one operator and offered for acceptance inspection at one time. Wherever practicable, the lot shall be identifiable in approximate order of production until sampling has been completed, by some method such as placing gaskets wound during each continuous period of 4 hours or less temporarily in separate containers.

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4.2.2 Sampling for visual and dimensional examination.- A random sample of gaskets shall be selected from each lot offered for Government examination in accordance with Standard MIL-STD-105 at inspection level III for lots of 40 and under, inspection level II for lots of 41 to 300 and inspection level I for lots of 301 and over. The Acceptable Quality Level (AQL) shall be 1.5 percent defective. However, the smallest sample size shall be 15. Where the lot is approximately identified in order of production, the sample gaskets shall be selected so that each period of continuous production is represented. Otherwise the sample shall be selected at random.

4.2.2.1 Sampling for ply count.- One third of the sample gaskets selected in accordance with 4.2.2 shall be subjected to the ply count in accordance with 4.3.1.1.

4.2.3 Sampling for tests.- One fifth of the sample gaskets selected in accordance with 4.2.2 shall be subjected to the lot acceptance test specified in 4.3.2. If the sample number should be odd, then the higher even number of sample gaskets shall be selected.

4.2.4 Sampling for comparison inspection.- The Government inspector shall select two sample gaskets for the inspection specified in 4.3.3 from the first lot of each type, series and size offered for inspection under each contract or order, and thereafter from one lot chosen from each group of 10 lots of each type, series and size offered for inspection under the contract or order which have passed the lot acceptance examination and tests at place of manufacture. Insofar as is practicable, the samples chosen for comparison inspection shall be duplicates of the samples chosen in accordance with 4.2.3. The samples shall be forwarded to the Naval Engineering Experiment Station, Annapolis, Maryland. In case lot acceptance tests are performed at the place of manufacture, a copy of the test data obtained from the samples chosen in accordance with 4.3.3 shall accompany the comparison inspection samples.

4.2.5 Sampling for determination of composition of metal strip.- A sample of approximately 2 ounces of the metal strip used in the manufacture of each lot of gaskets shall be taken before manufacturing begins and another 2-ounce sample shall be taken after the last one has been manufactured. Both samples shall be forwarded to a laboratory designated by the bureau or agency concerned for chemical analysis (see 4.5.4).

4.3 Examination and test.-

4.3.1 Visual and dimensional examination.- Each of the sample gaskets selected in accordance with 4.2.2 shall be visually and dimensionally examined to verify compliance with the requirements of this specification not involving tests. Any gasket in the sample containing one or more visual or dimensional defects shall be rejected and if the number of defective gaskets in any sample exceed the acceptance number for that sample, the lot represented by the sample shall be rejected.

4.3.1.1 Ply count.- The Government inspector shall carefully count the plies of the sample gaskets selected in accordance with 4.2.2.1. A magnifying glass or other suitable device shall be used to insure accuracy of the count. If any one of the samples is found not to conform to this specification, the entire lot shall be rejected.

4.3.2 Acceptance tests.-

4.3.2.1 Place of test.- Acceptance tests shall be performed at the place of manufacture if facilites are available. Otherwise, they shall be performed at the Engineering Experiment Station, Annapolis, Maryland.

4.3.2.2 Each of the sample gaskets, selected in accordance with 4.2.3, shall be tested in accordance with 4.5.3. If any sample gasket fails the test, the lot represented by the sample shall be rejected.

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4.3.3 Comparison inspection (at a Government laboratory).-

4.3.3.1 Procedure.- The sample gaskets selected and forwarded in accordance with 4.2.4 shall be subjected to the compression load test specified in 4.5.3 and to such other tests of those specified in 4.5 as the Government inspector or the Engineering Experiment Station considers necessary to establish conformance with this specification.

4.3.3.2 Action in case of failure.- Except as hereinafter specified, acceptance and rejection of lots shall normally be on the basis of the sampling, examination and tests of 4.2.2, 4.3.1 and 4.3.2 and acceptance shall not be withheld pending receipt of inspection reports on comparison inspection samples. However, upon receipt of an unsatisfactory inspection report on a comparison inspection sample, the Government inspector shall select additional samples from every subsequent lot offered for inspection which has passed the acceptance tests at place of manufacture. The additional samples shall be submitted to the Engineering Experiment Station, and shall there be subjected to the test or tests wherein failure was observed. Lots shall then be accepted only upon receipt of a satisfactory test report on the samples so selected. This additional testing shall be discontinued and lot acceptance returned to the normal basis when four successive lots have been accepted.

4.3.4 Rejected lots.- Rejected lots may be offered again for Government acceptance inspection provided the contractor has repair or removed all nonconforming gaskets. Samples shall again be selected from such resubmitted lots and inspected to verify compliance with this specification.

4.4 Control chart procedure.- For the information of the bureau or agency concerned, and to aid the manufacturer in controlling quality, the Government inspector, with the cooperation of the manufacturer, shall continuously maintain a control chart for compression test data for each type, series, and size of gaskets in accordance with the directions specified in 4.4.1 and 4.4.2. The compression thickness specified in 4.4.1 and 4.4.2 means the average thickness of the sample gasket at 25,000 p.s.i. bolt stress (root area) or 30,000 p.s.i. bolt stress (root area) as applicable (see tables I and II) when the sample is tested as specified in 4.5.3.

4.4.1 Directions for construction of the charts.- On suitable cross-section paper, a chart shall be laid out with vertical scale at the left for the mean of the two compressed thicknesses obtained from the pairs of sample gaskets tested in accordance with 4.3.2, and for the numerical value of the difference of the two compressed thicknesses, with a horizontal scale for date of sampling and inspection lot number. A horizontal central line shall be drawn for the mean of the compressed thickness value specified in tables I and II. Two horizontal control lines shall be drawn for the mean above and below the central line respectively at distances from the central line corresponding to a compressed thickness of 3.5 thousandths of an inch. One horizontal control line shall be drawn for the difference corresponding to 9.6 thousandths of an inch.

4.4.2 Directions for use of chart.- On the chart specified in 4.4.1, the successive means and differences of compressed thicknesses of pairs of gaskets tested in accordance with 4.3.2 shall be plotted. When comparison inspection on pairs of samples representing a lot are tested in accordance with 4.3.3, the points shall be plotted for the comparison inspection on the chart vertically in line with the points for the lot acceptance test, using some convenient device to distinguish the two types of points. The Government inspector shall inform the bureau or agency concerned and warn the manufacturer whenever a point falls outside control limits on the chart, and whenever a dangerous trend is observable in a succession of points, and (in case lot acceptance tests are performed at the place of manufacture) whenever the comparison inspection result plotted on the chart for means for all sizes lies either above or below all the corresponding ten lot acceptance points.

4.5 Test procedures.-

4.5.1 Line gasket endurance test.- The endurance test on type I line flange gasket and type II refills, shall be conducted in series 150 and 600 flanged joints of 6 inch line gasket test apparatus. This apparatus shall consist of a series of joints on 3 foot sections of piping. For series 150, gasket steam at 160 pounds gage pressure and 850°F. temperature shall be admitted to the apparatus at one end and

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discharged at the other end into a 150 pound gage pressure line. For series 300, 400 and 600, steam at 600 pounds gage pressure and 850°F. temperature shall be admitted to the apparatus at one end and discharged at the other end into a 590 pound gage pressure line. The piping sections shall be thoroughly insulated in accordance with the requirements for shipboard installation. The joints shall be examined for leakage three times daily. At the end of each working day the valves in the steam inlet lines to the apparatus shall be closed, the pressure within the apparatus shall be reduced, and boiler feed water reduced to 75 p.s.i. pressure and 140°F. temperature for series 150, and 300 p.s.i. and 140°F. for series 300, 400 and 600 shall be quickly but safely admitted to the apparatus to cause mechanical shocks and vibrations. Admission of the boiler feed water shall be continued until the apparatus becomes full; all valves then shall be closed; the flange covers shall be removed and the apparatus allowed to cool overnight and weekends. The test shall be continued for a period of 1,200 hours of superheated steam and boiler feed water conditions. Sample gaskets shall be tested for thermal shock as follows: With the apparatus in assembled cold condition, admit steam to the apparatus at pressures and temperatures corresponding to the ratings of all series. This test shall be accomplished altogether 20 times.

4.5.2 Line thermal shock test (type I).- The apparatus used for this test shall consist of 6 inch size pipe sections flanged together with welding-neck flanges to provide two joints. The center or spool-piece section shall consist of two flanges butt welded together. The end sections shall consist of 6 inch size pipe flange at one end and blanked off at the other. One end section shall be provided with steam and boiler feedwater inlet connections and the other with an exhaust connection. Two nozzles for directing a spray of boiler feedwater on the inner periphery of the test joints shall be centrally located within the apparatus, one at the centerline of each joint. Each nozzle shall consist of a 1/2 inch size pipe with 6 holes of 1/8 inch diameter drilled radially at 60 degree intervals around the pipe. Thermocouples shall be placed appropriately for measurement of temperatures at outer surface of pipe, at the raised faces of the test joints. of inlet steam and boiler feedwater, and at the exhaust connection. The test shall consist of 20 cycles of thermal shock obtained by changing conditions from steam at 600 p.s.i.g. and 850°F. to boiler feedwater at 700 p.s.i.g. and 250°F. The test joints shall be inspected closely for leakage during the shocks.

4.5.3 Compression load test.-

4.5.3.1 The test shall consist of subjecting the gaskets to compression in a Baldwin-Southmark, or equal, 600,000 pound hydraulic compression testing machine and measuring the thickness of the gaskets while under the test loads specified in tables I and II.

4.5.3.2 The gaskets shall be tested between steel test plates, the surfaces of which shall have a smooth finish, machined with a circular lay (concentric or spiral) having 60 to 100 serrations per inch of face width with depth of serrations not exceeding 0.003 inch or having an RHR finish of 60 to 1000. For special installations involving radioactive service or hazardous fluids where a finer finish is required, a maximum RHR of 500 shall be specified. The test plates shall be centrally located under the ball and socket head of the test machine. The ball and socket head is used to equally distribute the load applied to the gasket. By means of inverted T strap the upper compression plate is suspended and held in place when inserting a gasket. Four dial indicators graduated in 1/1000 inch divisions located 90 degrees apart and mounted on the upper compression plate shall be employed for measuring the thickness of the gasket.

4.5.3.3 During the test, the increment of load shall be applied until a load corresponding to 25,000 or 30,000 p.s.i. bolt stress (root area) is obtained, as applicable, for each specific size of gasket in accordance with tables I and II. The maximum rate of loading shall be 2,000 pounds per minute (p.p.m.). The dial indicators shall be set so that a zero reading is obtained with the maximum travel of the indicator plunger. The plates shall be then separated, a gasket centrally located between plates and the upper plate brought into contact with the gasket but no load applied. The dial indicators shall then be read, and the average of the four readings shall closely check the original micrometer thickness of the gasket. The load shall then be applied and the rate of loading depends on the size of gasket and the maximum load to be applied. At each interval, the specified load is maintained only for a sufficient time, 5 to 10 seconds, to obtain the dial indicator readings. By means of reading the dial indicator scales counterclockwise instead of clockwise, the actual thickness of the gasket is directly obtained. The average of the four dial indicator readings is considered to be the thickness of the gasket. After completion of the test, the load shall be

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released but the upper compression plate shall be left in contact with the gasket for approximately 10 seconds and the dial indicator readings shall be taken to determine the recovery of the gaskets. The test plates shall then be separated, the gasket removed from the test machine, measured for thickness by micrometers, examined for buckling on the inner periphery, condition of welds, and binding of the gasket filler in the outer metal ring. It may be noted that the final micrometer thickness and the released average dial indicator thickness of the gasket will be approximately the same providing the dial indicators are in proper adjustment. Occasionally between test runs, the zero setting of the dial indicators is checked with the test plates in contact with one another. No buckling on the inner edge of the gasket will be allowed. Gaskets shall at least recover to 0.145 inch minimum after the compression test of 4.5.3.3.

4.5.3.4 In order to test the gaskets ranging in size from 1/4 to 20 inches, inclusive, approximately the following size plates shall be used:

4.5.3.4.1 For gasket 1/4 to 2-1/2 inches, inclusive, the lower compression plate is 4-1/2 inches in diameter and 2-1/2 inches thick. The upper compression plate is 4-1/2 inches in diameter, 3 inches thick and is suspended from the frame.

4.5.3.4.2 For gaskets 3 to 6-1/2 inches, inclusive, the lower plate is 9 inches in diameter and 2-1/2 inches thick. The upper compression plate is 9 inches in diameter, 3 inches thick and suspended from the frame.

4.5.3.4.3 For gaskets 7 to 11-1/2 inches, inclusive, the lower plate is 14-1/2 inches in diameter, 3 inches thick and is suspended from the frame. The upper compression plate is 19 inches in diameter 1 inch thick and is suspended from the frame by means of the inverted T straps.

4.5.3.4.4 For gaskets larger than 11-1/2 inches, the lower plate is 19 inches in diameter and 2-1/2 inches thick. The upper compression plate is 19 inches in diameter, 3 inches thick, and is suspended from the frame.

4.5.3.4.5 Gaskets may be tested on plates having dimensions which differ from those given in 4.5.3.4.1 to 4.5.3.4.4, inclusive, provided the loading and deflections are within the limits of tables I and II, and provided that sample gaskets so tested will meet the requirements of this specification when tested at the Naval Engineering Experiment Station, Annapolis, Maryland.

4.5.4 Chemical analysis.- The sample selected in accordance with 4.2.5 shall be subjected to a chemical analysis to determine conformance with 3.3.1.

4.6 Inspection procedures.- For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.2).-

5.1.1 Level A.- Gaskets shall be packaged in accordance with method III of Specification MIL-P-116. The gaskets shall be packaged in quantities of 10 in containers conforming to Specification PPP-B-566, PPP-B-676, MIL-B-4229, or PPP-B-636, at the option of the contractor. Box closures shall conform to the applicable box specification and the appendix thereto. The gross weight shall not exceed the limitations of the applicable box specification.

5.1.2 Level C.- Gaskets shall be preserved and packaged in accordance with the manufacturer's commercial practice.

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5.2 Packing (see 6.2).-

5.2.1 Level A.- Gaskets, packaged as specified (see 6.2) shall be packed in overseas type wood cleated fiberboard, nailed wood, wirebound wood, corrugated or solid fiber, wood cleated paper overlaid, or wood cleated plywood boxes conforming to Specification PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-636 (class 3), MIL-B-10377 or PPP-B-601, respectively, at the option of the contractor. Shipping containers shall have caseliners conforming to Specification MIL-L-10547 and shall be closed and sealed in accordance with the appendix thereto. Caseliners for boxes conforming to Specification PPP-B-636 may be omitted provided all joints and corners of the boxes are sealed with minimum 1-1/2 inch wide sensitive tape conforming to Specification PPP-T-76. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 Level B.- Gaskets, packaged as specified (see 6.2) shall be packed in domestic type wood cleated fiberboard, nailed wood, wirebound wood, cleated plywood or wood cleated paper overlaid boxes or class 2 fiber boxes conforming to Specification PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-601, MIL-B-10377, or PPP-B-636, respectively, at the option of the contractor. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of wood boxes shall not exceed 200 pounds; fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 Level C.- Gaskets, packaged as specified (see 6.2) shall be packed in a manner which will insure acceptance and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.3 Marking.- In addition to any special marking required by the contract or order or herein, interior and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use.- Type II refills are intended to be used with existing rings on board ships and type I gaskets are not to be ordered except where rings are not available.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, series and size of gaskets required, and type of flanges with which gaskets are to be used (see 1.2).
(c) For type II gaskets, the inside and outside dimensions (see 3.2.2).
(d) For type I gaskets, the marking required on metal ring (see 3.5).
(e) Level of packaging and packing required (see 5.1 and 5.2).

6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-16265, whether or not such products have actually been so listed by that date.

6.3.1 The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Department of the Navy, Washington, 25, D.C.

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6.4 Superseding data.- This specification replaces that part of Specification MIL-G-16265A covering classes b (in part) and c, as follows:

MIL-G-16265A	MIL-G-0016265B(SHIPS)
Class b (in part)	Type I
Class c	Type II

Classes a and b (in part) of Specification MIL-G-16265A have been replaced by types I and II of Specification MIL-G-21032 and gaskets for commercial flanges should be purchased under Specification MIL-G-21032.

Notice.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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